B2. Wonderful Coloring - 2

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: binary search, constructive algorithms, data structures, greedy |
https://codeforces.com/problemset/problem/1551/B2

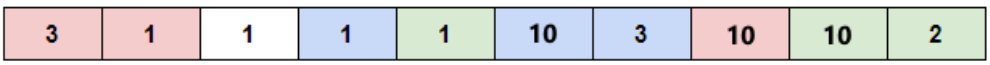
This problem is an extension of the problem "Wonderful Coloring - 1". It has quite many differences, so you should read this statement completely.

Recently, Paul and Mary have found a new favorite sequence of integers $a_1, a_2, \ldots, a_n$. They want to paint it using pieces of chalk of k colors. The coloring of a sequence is called *wonderful* if the following conditions are met:

1. each element of the sequence is either painted in one of k colors or isn't painted;
2. each two elements which are painted in the same color are different (i. e. there's no two equal values painted in the same color);
3. let's calculate for each of k colors the number of elements painted in the color — all calculated numbers must be equal;
4. the total number of painted elements of the sequence is the **maximum** among all colorings of the sequence which meet the first three conditions.

E. g. consider a sequence a=[3, 1, 1, 1, 1, 10, 3, 10, 10, 2] and k=3. One of the wonderful colorings of the sequence is shown in the figure.

The example of a wonderful coloring of the sequence a=[3, 1, 1, 1, 1, 10, 3, 10, 10, 2] and k=3. Note that one of the elements isn't painted.



Help Paul and Mary to find a wonderful coloring of a given sequence a.

Input

The first line contains one integer t ($1 \le t \le 10000$) — the number of test cases. Then t test cases follow.

Each test case consists of two lines. The first one contains two integers n and k ($1 \le n \le 2 \cdot 10^5$, $1 \le k \le n$) — the length of a given sequence and the number of colors, respectively. The second one contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le n$).

It is guaranteed that the sum of n over all test cases doesn't exceed $2 \cdot 10^5$.

Output

Output t lines, each of them must contain a description of a wonderful coloring for the corresponding test case.

Each wonderful coloring must be printed as a sequence of n integers $c_1, c_2, \ldots, c_n$ ($0 \le c_i \le k$) separated by spaces where

- c_i=0, if i-th element isn't painted;

- c_i gt 0, if i-th element is painted in the c_i-th color.

Remember that you need to maximize the total count of painted elements for the wonderful coloring. If there are multiple solutions, print any one.

Example

Input

```
6
10 3
3 1 1 1 1 10 3 10 10 2
4 4
1 1 1 1
1 1
1
13 1
3 1 4 1 5 9 2 6 5 3 5 8 9
13 2
3 1 4 1 5 9 2 6 5 3 5 8 9
13 3
3 1 4 1 5 9 2 6 5 3 5 8 9
```

Output

```
1 1 0 2 3 2 2 1 3 3
4 2 1 3
1
0 0 1 1 0 1 1 1 0 1 1 1 0
2 1 2 2 1 1 1 1 2 1 0 2 2
1 1 3 2 1 3 3 1 2 2 3 2 0
```

Note

In the first test case, the answer is shown in the figure in the statement. The red color has number 1, the blue color — 2, the green — 3.